* A linear operator T is said to be diagonalisable if $\exists$ a basis S of a vector space V such that T is represented by a diagonal matrix.

The basis S is said to diagonalise T.

Theorem:-

Let A be the matrix representation of a linear operator T. then T is diagonalisable iff $\exists$ an invertible matrix P such that $P^{-1}AP = D$

i.e. T is diagonalisable iff its matrix representation can be diagonalised by a similarity transformation.

Q:- Suppose $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & -4 \\ 1 & -2 & 2 \end{bmatrix}$

Find a matrix B that represents the linear operator A relative to the basis $S = \{(1,1,0)^T, (0,1,1)^T, (1,2,2)^T\}$

$$\begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & -4 \\ 1 & -2 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 4 & 9 \\ 7 & 1 & 4 \\ -1 & 0 & 1 \end{bmatrix}$$

~~$(a,b,c) = \alpha(1,1,0) + \beta(0,1,1) + \gamma(1,2,2)$~~

$= \alpha+\gamma$, $[A(u_1)]_S$ $[A(u_2)]_S$ $[A(u_3)]_S$.

$[A(u_1)]_S = v_1$, $[A(u_2)]_S = v_2$, $[A(u_3)]_S = v_3$

Another method

$E = \{e_1, e_2, e_3\}$

Change of basis $P \to E$ to S

$$[B] = P^{-1}AP.$$

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